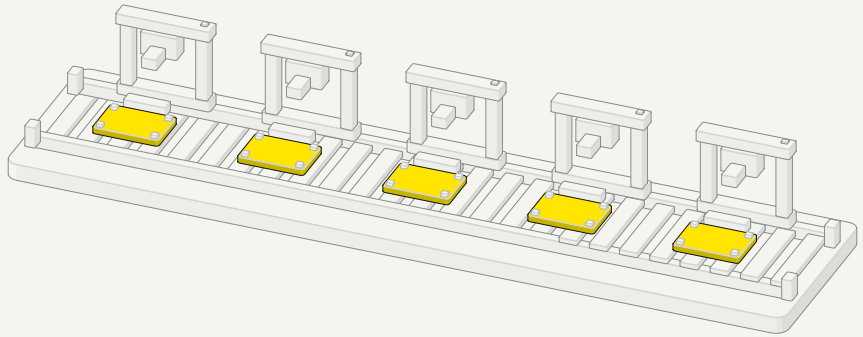
A CORE YOU CAN EXPLAIN

An educational five-stage RV32I core with DOT4.
Review deck for the October 17–18 build sprint.



Bits carry an interpretation

0x80 means 128 unsigned or −128 as a signed byte.

Sign extension preserves the signed value.

Bits	Unsigned 8-bit	Signed 8-bit	32-bit sign extension
01111111	127	127	0x0000007F
10000000	128	−128	0xFFFFFF80
11111111	255	−1	0xFFFFFFFF

Illustrations explain the mechanism. Measured figures carry their stated scope.

Clocked RTL preserves old state

Nonblocking assignments update registers after evaluation.
Combinational logic computes the next value between edges.

FIG 02 / REGISTER BOUNDARY

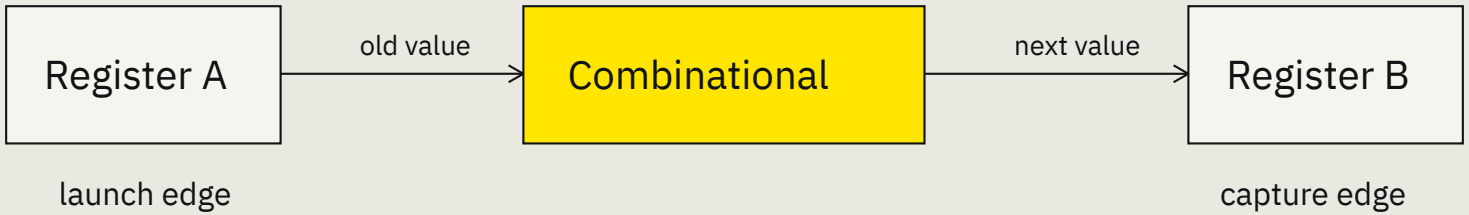


Illustration: combinational logic settles between active edges.

Illustrations explain the mechanism. Measured figures carry their stated scope.

The instruction contract

Decode identifies the operation, source registers and destination.
The ISA defines results. The pipeline chooses their timing.

FIG 03 / R-TYPE FIELD MAP

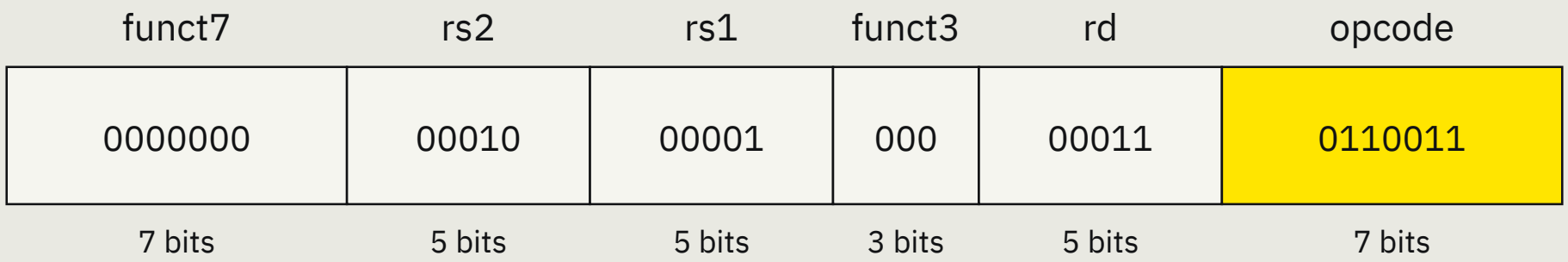


Illustration: field widths sum to 32 bits. Widths use a shared bit scale.

Illustrations explain the mechanism. Measured figures carry their stated scope.

Five stages overlap instructions

Each valid instruction advances through IF, ID, EX, MEM and WB.
Ten ideal independent instructions need fourteen cycles.

FIG 05A / LOGICAL STAGES



Illustration: stage boundaries hold the instruction and its control state.

Illustrations explain the mechanism. Measured figures carry their stated scope.

Forwarding priority

The newest matching available producer wins.

MEM non-load result has priority over WB. x0 never produces a value.

FIG 06 / EX OPERAND SELECTION

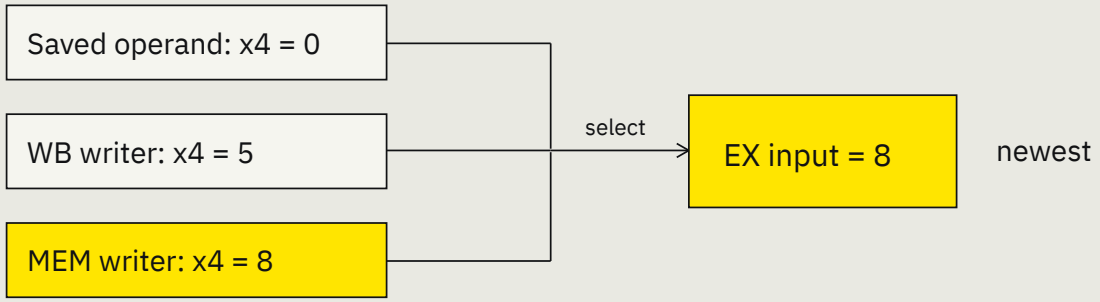


Illustration: the matching non-load MEM result has priority over WB.

The load-use bubble

Hold PC and IF/ID. Insert an invalid EX entry.
The load advances, then WB supplies its data to the consumer.

At stall edge	Action
PC	Hold current fetch address
IF/ID	Hold consumer instruction
ID/EX	Insert invalid bubble
EX/MEM and MEM/WB	Older work continues

Illustrations explain the mechanism. Measured figures carry their stated scope.

A branch resolves in EX

A taken branch discards the two younger front-end stages.
Wrong-path stores must never reach a memory side effect.

FIG 08 / EX-STAGE REDIRECT

Instruction	1	2	3	4	5	6
branch	IF	ID	EX	MEM	WB	
wrong-path store		IF	ID / discard			
wrong-path ADDI			IF / discard			
target				IF	ID	EX

Illustration: the branch discards younger IF and ID instructions at the end of cycle 3.

DOT4 uses signed byte lanes

Four independent products feed one signed sum.
It has no accumulator input or saturation.

FIG 09 / DOT4 DATAFLOW

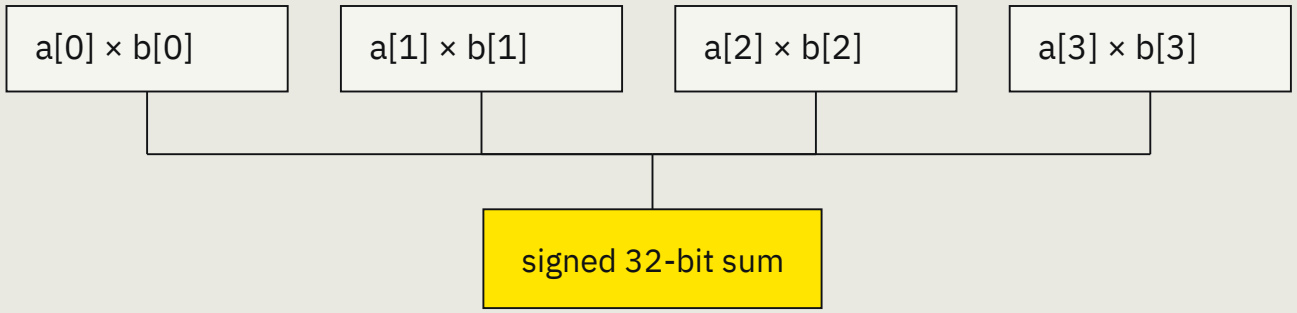


Illustration: independent signed-byte products feed one sum.

Illustrations explain the mechanism. Measured figures carry their stated scope.

One worked DOT4 example

A=0x0102FF80, B=0x020304FF.

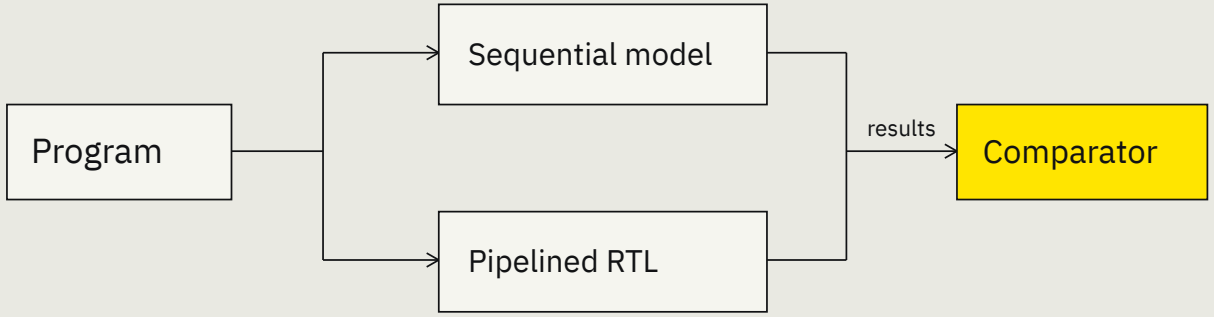
128 – 4 + 6 + 2 = 132.

Lane	A byte	A signed	B signed	Product
0	0x80	−128	−1	128
1	0xFF	−1	4	−4
2	0x02	2	3	6
3	0x01	1	2	2
Total				132

Verification compares behavior

The local checker compares ordered retirements and every byte write.
Hand calculations and external checks strengthen the evidence.

FIG 10 / TWO EXECUTIONS, ONE COMPARISON



Same program words drive both implementations. A mismatch identifies the first divergent result.

Coverage has a narrow scope

Reference measurements. A passing bin plan does not cover every hazard.
The compact core has only 73 line instrumentation points.

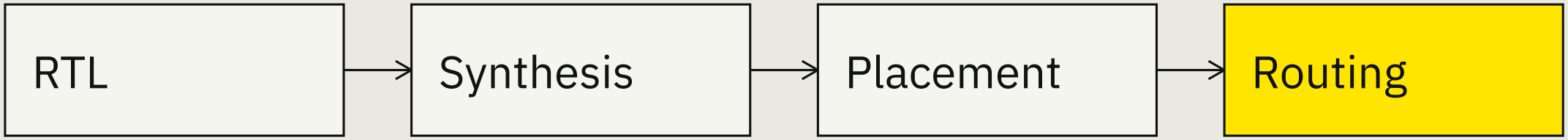
Metric	Reference result	Scope
Passing programs	253 / 253	Directed, random and hazard suite
Retired instructions	11,928 checked	Ordered architectural trace
Line instrumentation	73 / 73 = 100.00%	Core only, Verilator 5.020
Functional bins	46 / 46	Named class/outcome plan
Hazard cross	327 / 327	Producer × consumer × distance

Illustrations explain the mechanism. Measured figures carry their stated scope.

FPGA implementation evidence

36.05 MHz median internal Fmax estimate, seeds 1–5. Constraint: 12 MHz.
5,539 HX8K logic cells. Memories, peripherals and board operation excluded.

FIG 11 / IMPLEMENTATION FLOW



The final timing estimate belongs to a device, constraints, tools and route seed.

Illustrations explain the mechanism. Measured figures carry their stated scope.

Tomorrow’s completion gate

Explain a dependency trace, a DOT4 result and the route report.

Publish your tests and experiment notes. Reproduce from a clean clone.

Illustrations explain the mechanism. Measured figures carry their stated scope.